

NEXORA MediTwin: Technical & Clinical Documentation (2025)



1. Executive Introduction

MediTwin is an advanced AI-powered healthcare ecosystem designed to bridge the gap between biological signals and predictive medicine. By creating a personalized Digital Twin in the cloud, we enable continuous vital monitoring and early detection of health risks.

2. System Architecture & Hardware

The system is built on a high-performance IoT backbone.

Microcontroller: ESP32 Dual-Core Processor for simultaneous data collection and transmission.

Sensing Suite:

MAX30102: High-sensitivity pulse oximetry and heart rate sensor.

AD8232: Single-lead heart rate monitor for high-fidelity ECG waveforms.

Communication Protocol: I2C for internal sensor communication and secure WebSockets for real-time cloud streaming.

Sampling Rate: 500Hz high-frequency sampling for precise morphological analysis.

3. The AI Neural Backbone

MediTwin utilizes a proprietary multi-layer AI architecture to interpret biology.

Layer 1 (CNN Analyzers): Processes ECG data to identify arrhythmias and morphological anomalies.

Layer 2 (TinyML Edge AI): Lightweight Random Forest models running on-chip for offline fall detection and critical alerts.

Layer 3 (LSTM Forecasts): Predicts vital sign trajectories for the next 48–72 hours based on historical baselines.

4. Clinical Validation Parameters

Our models achieve industry-leading accuracy for proactive care.

Vital Monitoring: Real-time tracking of Heart Rate (BPM), SpO2 (%), Temperature (°C), and Stress Levels (GSR).

5. User Setup & Deployment

Hardware Application

Preparation: Clean the upper left chest area with an alcohol wipe.

Application: Press the Bio-Patch firmly against the skin to ensure the bio-sensors make consistent contact.

Activation: Hold the primary power button for 3 seconds until the "Bio-Sensor Active" light flashes.

Dashboard Synchronization

Navigate to your secure MediTwin Dashboard.

The system will automatically recognize your device ID and begin real-time visualization.

6. Troubleshooting & Maintenance

Waiting for Data: Ensure the device is within 10 meters of your internet gateway.

Poor Signal: Check patch placement; ensure the AD8232 sensor is not obstructed by clothing or debris.

Power Failure: Charge the patch using the standard USB-C interface for at least 2 hours.

7. Privacy & Security Compliance

Encryption: All data is encrypted at the hardware level before transmission.

Validation: Data integrity is maintained via Pydantic-enforced schema validation in the FastAPI backend.